

content...

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import numpy as np

def incmatrix(genl1,genl2):
m = len(genl1)
  n = len(genl2)
  M = None #to become the incidence matrix
  VT = np.zeros((n*m,1), int) #dummy variable

#compute the bitwise xor matrix
M1 = bitxormatrix(genl1)
M2 = np.triu(bitxormatrix(genl2),1)

for i in range(m-1):
  for j in range(i+1, m):
    [r,c] = np.where(M2 == M1[i,j])
    for k in range(len(r)):
      VT[(i)*n + r[k]] = 1;
      VT[(i)*n + c[k]] = 1;
      VT[(j)*n + r[k]] = 1;
      VT[(j)*n + c[k]] = 1;

    if M is None:
      M = np.copy(VT)
    else:
      M = np.concatenate((M, VT), 1)

  VT = np.zeros((n*m,1), int)

return M
```